

J Guarantee for UncertainStateAction

In this section, we present the main guarantee of `UncertainStateAction` (Algorithm 6) as a standalone algorithm; see Lemma J.1. Then, in Lemma J.2, we provide its guarantee when used as a subroutine within MTSS (Algorithm 4). For a discussion of the motivation for these results, we refer back to Section I.1.3.

Lemma J.1. *Consider a call to `UncertainStateActionh`($\mathcal{C}_{0:h-1}, \hat{\pi}_{1:h}, \Sigma_h; \mathbf{a}, N, \bar{N}$) (Algorithm 6) for some given $h, \mathcal{C}_{0:h-1}, \hat{\pi}_{1:h}, \Sigma_h, \mathbf{a} \in \mathcal{A}$, N , and $\bar{N}$ such that $\sigma_{\min}(\Sigma_h) \geq \lambda$, for some $\lambda \in (0, 1)$. Then, for any $\delta' \in (0, 1)$ and $\zeta \in (0, 1/2)$, with probability at least $1 - \delta'$, the output $(\hat{x}_h, \hat{a}_h)$ of `UncertainStateAction` satisfies:*

- For all $\ell \in [0..h-1]$ and $(x_\ell, a_\ell) \in \mathcal{C}_\ell$,

$$\mathbb{P}_{\hat{\pi}_{\ell+1:h}} \left[\|\varphi_h(\mathbf{x}_h, \mathbf{a}_h)\|_{\Sigma_h^{-1}}^2 > 2 \left(\zeta \vee \|\varphi_h(\hat{x}_h, \hat{a}_h)\|_{\Sigma_h^{-1}}^2 \right) \mid \mathbf{x}_\ell = x_\ell, \mathbf{a}_\ell = a_\ell \right] \leq \max_{h \in [H]} \frac{4 \log \left(\frac{16H|\mathcal{C}_h|}{\lambda \delta' \zeta} \right)}{N}, \quad (90)$$

where $\varphi_h(\cdot, \cdot) := \phi_h(\cdot, \cdot) - \phi_h(\cdot, \mathbf{a})$.

- Furthermore, there exists $\mathcal{X}_{h,\text{span}} \subseteq \mathcal{X}$ such that for all $\ell \in [0..h-1]$ and $(x_\ell, a_\ell) \in \mathcal{C}_\ell$, $\mathbb{P}_{\hat{\pi}_{\ell+1:h-1}}[\mathbf{x}_h \in \mathcal{X}_{h,\text{span}} \mid \mathbf{x}_\ell = x_\ell, \mathbf{a}_\ell = a_\ell] \geq 1 - \max_{h \in [H]} \frac{4}{N} \log \frac{32HN|\mathcal{C}_h|}{\lambda \delta' \zeta}$ and

$$\forall x_h \in \mathcal{X}_{h,\text{span}}, \quad \mathbb{P}_{\mathbf{a} \sim \pi_{h,\text{ref}}(\cdot \mid x_h)} \left[\|\varphi_h(x_h, \mathbf{a})\|_{\Sigma_h^{-1}}^2 > 2 \left(\zeta \vee \|\varphi_h(\hat{x}_h, \hat{a}_h)\|_{\Sigma_h^{-1}}^2 \right) \right] \leq \max_{h \in [H]} \frac{4 \log \left(\frac{16H|\mathcal{C}_h|}{\lambda \delta' \zeta} \right)}{N}. \quad (91)$$

Proof of Lemma J.1. Fix $\delta' \in (0, 1)$ and $\zeta \in (0, 1/2)$, and let $\Gamma := \{\zeta, 2\zeta, \dots, \lceil \frac{4}{\zeta \lambda} \rceil \zeta\}$. Further, for $\ell \in [0..h-1]$ and $(x_\ell, a_\ell) \in \mathcal{C}_\ell$, let $\mathcal{D}_\ell(x_\ell, a_\ell)$ be the dataset in Algorithm 6 when the algorithm returns. Note that $\mathcal{D}_\ell(x_\ell, a_\ell)$ consists of N i.i.d. pairs sampled from $\mathbb{P}_{\hat{\pi}_{\ell+1:h}}[(\mathbf{x}_h, \mathbf{a}_h) = \cdot \mid \mathbf{x}_\ell = x_\ell, \mathbf{a}_\ell = a_\ell]$. Thus, by Freedman's inequality (Lemma C.2) and the union bound over $\ell \in [0..h-1]$, $(x_\ell, a_\ell) \in \mathcal{C}_\ell$, and $\gamma \in \Gamma$, there is an event $\mathcal{E}$ of probability at least $1 - \delta'/2$ under which,

$$\begin{aligned} \forall \ell \in [0..h-1], \forall (x_\ell, a_\ell) \in \mathcal{C}_\ell, \forall \gamma \in \Gamma, \quad & \mathbb{P}_{\hat{\pi}_{\ell+1:h}} \left[\|\varphi_h(\mathbf{x}_h, \mathbf{a}_h)\|_{\Sigma_h^{-1}}^2 \geq \gamma \mid \mathbf{x}_\ell = x_\ell, \mathbf{a}_\ell = a_\ell \right] \\ & \leq \frac{2}{N} \sum_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \mathbb{I} \left\{ \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 > \gamma \right\} + \frac{4 \log(2H|\mathcal{C}_\ell|/\delta')}{N}, \\ & \leq \frac{2}{N} \sum_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \mathbb{I} \left\{ \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 > \gamma \right\} + \frac{4 \log \left(\frac{16H|\mathcal{C}_\ell|}{\lambda \delta' \zeta} \right)}{N}, \end{aligned} \quad (92)$$

where the last step follows by the facts that $|\Gamma| \leq \lceil \frac{4}{\zeta \lambda} \rceil$, $\lambda \in (0, 1)$, and $\zeta \in (0, 1/2)$. Now, since $\sigma_{\min}(\Sigma_h) \geq \lambda$ and $\sup_{(x,a) \in \mathcal{X} \times \mathcal{A}} \|\phi_h(\cdot, \cdot)\| \leq 1$ (Assumption H.1), we have that $\sup_{(x,a) \in \mathcal{X} \times \mathcal{A}} \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 \leq \frac{4}{\lambda}$. Therefore, by the definition of Γ , we have that for all $\ell \in [0..h-1]$ and $(x_\ell, a_\ell) \in \mathcal{C}_\ell$, there exists $\gamma_\ell(x_\ell, a_\ell) \in \Gamma$ such that

$$\begin{aligned} \max_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 & \leq \gamma_\ell(x_\ell, a_\ell), \\ & \leq \max_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 + \zeta, \\ & \leq 2 \left(\zeta \vee \max_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 \right), \\ & \leq 2 \left(\zeta \vee \|\varphi_h(\hat{x}_h, \hat{a}_h)\|_{\Sigma_h^{-1}}^2 \right), \end{aligned} \quad (93)$$

where the last inequality follows by the fact that

$$\max_{\ell \in [0..h-1]} \max_{(x,a) \in \mathcal{D}_\ell(x_\ell, a_\ell)} \|\varphi_h(x, a)\|_{\Sigma_h^{-1}}^2 \leq \|\varphi_h(\hat{x}_h, \hat{a}_h)\|_{\Sigma_h^{-1}}^2$$